

SHAMANTH BIDARE RAMESH

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EDUCATION

Malnad College Of Engineering, Hassan, Karnataka, India

2026

PROFILE

Information Science student at Malnad College of Engineering and Generative AI Intern at AnalogData. Specialized in Small Language Models (SLM), Agentic AI, and RAG architectures. Proven track-record in graphic designing, frontend web development, with a focus on building autonomous systems and deep learning applications.

PROFESSIONAL EXPERIENCE

AnalogData, India : Generative AI (SLM & Agentic AI) Intern

Researching and implementing Small Language Models (SLMs) and Agentic AI frameworks to optimize local AI deployments, achieving high-performance inference on edge devices with limited compute. Developing multi-agent autonomous workflows to automate complex data extraction and processing tasks, significantly reducing manual intervention and operational latency. Architecting Retrieval-Augmented Generation (RAG) pipelines tailored for SLMs to improve factual accuracy and reduce hallucinations in domain-specific applications. Optimizing local model execution using tools like Ollama and Qdrant techniques, chromadb to ensure privacy-first, offline AI capabilities for enterprise use cases. Evaluating and benchmarking model performance across various Agentic frameworks (like Langchain, LangGraph, llamaindex, etc) to determine the most stable architectures for production-level agents.

PROJECTS

FinGenie – AI-Powered Financial Literacy Platform

Architected an end-to-end educational platform designed to teach financial literacy to children through interactive, AI-driven guidance and personalized learning paths. Implemented Gamification Modules (Progress tracking, achievements, and rewards) to increase student engagement and retention, translating dry financial concepts into interactive challenges. Engineered a Secure Backend API using FastAPI to manage user data, progress logs, and seamless integration with Small Language Models (SLMs) for real-time tutoring. Designed a Simplified UI/UX specifically for younger demographics, focusing on visual cues and intuitive navigation to make complex topics like budgeting and saving accessible.

EcoTrack: Smart E-Waste Management System

Engineered a comprehensive IoT and Web-based solution to streamline the lifecycle of electronic waste, from consumer disposal to certified recycling centers. Developed an automated classification module (potentially using CNNs or Image Recognition) to identify and categorize different types of electronic components for optimal recycling. Architectured the system for scalability, ensuring the backend could handle high-concurrency requests from multiple geo-locations during peak disposal periods.

Generative AI & Local RAG Developer

Developed a Local RAG (Retrieval-Augmented Generation) system utilizing LangChain to orchestrate a seamless pipeline between private data and Large Language Models. Integrated Ollama 3.2 for high-performance, local LLM inference. Architectured a high-dimensional vector storage system using ChromaDB, implementing Chroma's native embedding functions to manage and query document embeddings with high retrieval precision. Optimized data ingestion by implementing advanced text-splitting strategies (e.g., RecursiveCharacterTextSplitter), significantly improving the model's context window efficiency and reducing response "hallucinations." Tech Stack: Python, LangChain, Ollama (Llama 3.2), ChromaDB, Vector Embeddings.

SKILLS

AI/ML: Agentic AI, Small Language Models (SLMs), RAG

Tools & Frameworks: FastAPI, Docker, Kubernetes, Django, MCP, Langchain, LangGraph, LangServer, LangSmith, Ollama, Git, Github, canave, Figma, AWS

Programming Languages & OS: Python, UI/UX, Windows, Linux(Ubuntu)

Databases & other: MySQL, MongoDB, chromadb & Canva, Figma, capcut

ACTIVITIES AND ACHIEVEMENTS

Secured 1st Place in Techathon'23 at SRIT Coimbatore, Tamil Nadu • 2nd Place in Paper 2 Perspective event at Innofesta, HITAM College, Hyderabad • 3rd Place at IIT Bombay in the Kohii, Techfest event

Certification : AWS Academy Cloud Architecting and completed the Kaggle x Google AI Agents

Paper Presenter: ICAIES 2025 – Presented research on Artificial Intelligence in Engineering Systems.